

13/07/20

S T Q Q S S D

→ A2M8 - Lista

QUESTÃO	1	2	3	4	5	6	7	II	III	IV
RESPOSTA	C	C	D	E	C	D	D	A	C	B

① Encontrar as derivadas parciais em relação a todos as variáveis da função.

① $f(x, y) = 5x^7 y$

$$\frac{dz}{dx} = (5x^7) y = \boxed{35x^6 y}, \quad \frac{dz}{dy} = \boxed{5x^7} \quad \text{ALTERNATIVA C}$$

② $f(x, y) = (x^2 + y^2)^{\frac{3}{2}}$

$$\frac{dz}{dx} \Rightarrow u = x^2 + y^2 \rightarrow (u)^{\frac{3}{2}} \rightarrow \frac{du}{dx} = \frac{3}{2} u^{\frac{1}{2}} \rightarrow \frac{3}{2} u^{\frac{1}{2}} \text{ ou } \frac{3}{2} (x^2 + y^2)^{\frac{1}{2}} \cdot 2x$$

$$\Rightarrow 3x(x^2 + y^2)^{\frac{1}{2}} \text{ ou } \boxed{3x \sqrt{x^2 + y^2}}$$

$$\frac{dz}{dy} = \boxed{3y \sqrt{x^2 + y^2}} \quad (\text{mesma coisa}) \quad \text{ALTERNATIVA C}$$

③ $f(x, y) = 60x^{\frac{3}{4}} y^{\frac{1}{4}}$

$$\frac{dz}{dx} = x^{\frac{3}{4}} (60 y^{\frac{1}{4}}) = \frac{3}{4} x^{-\frac{1}{4}} \cdot 60 y^{\frac{1}{4}} = \boxed{45 x^{-\frac{1}{4}} y^{\frac{1}{4}}}$$

$$\frac{dz}{dy} = y^{\frac{1}{4}} (60 x^{\frac{3}{4}}) = \frac{1}{4} y^{-\frac{3}{4}} 60 x^{\frac{3}{4}} = \boxed{15 y^{-\frac{3}{4}} x^{\frac{3}{4}}}$$

ALTERNATIVA D

$$(4) f(x, y) = x^2 y^4 + \sin(y)$$

$$\frac{dz}{dx} = \boxed{2xy^4 + \sin(y)}$$

ALTERNATIVA E

$$\frac{dz}{dy} = \boxed{4y^3 x^2 + \cos(y)}$$

$$(5) f(x, y, z) = \frac{x^2}{24} + \frac{y^2}{12} + \frac{z^2}{6} - 1$$

$$\frac{df}{dx} = \frac{2x}{24} = \boxed{\frac{x}{12}}$$

ALTERNATIVA C

$$\frac{df}{dy} = \frac{2y}{12} = \boxed{\frac{y}{6}}$$

$$\frac{df}{dz} = \frac{2z}{6} = \boxed{\frac{z}{3}}$$

$$(6) f(x, y) = \frac{x^2}{y} - \frac{y}{x^2}$$

$$\frac{dz}{dx} = \frac{2x}{y} - \frac{y}{x^3} = \frac{2x}{y} - yx^{-3} \Rightarrow \frac{2x}{y} + 2yx^{-3} = \boxed{\frac{2x}{y} + 2yx^{-3}}$$

$$\frac{dz}{dy} = x^2 y^{-1} - yx^{-2} = -x^2 y^{-2} - x^{-2} \text{ ou } \boxed{\frac{-x^2}{y^2} - \frac{1}{x^2}}$$

ALTERNATIVA D

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$$\textcircled{7} f(x, y) = e^{2x} \sin(y)$$

$$\frac{dz}{dx} = \boxed{2e^{2x} \sin(y)}$$

(Integral imediata)

ALTERNATIVA D.

$$\frac{dz}{dy} = \boxed{e^{2x} \cos(y)}$$

(Integral imediata)

III Ache a equação polar equivalente à " $x^2 + y^2 - 4y = 0$ ".

$$(r \cos(\theta))^2 + (r \sin(\theta))^2 - 4r \sin(\theta) = 0$$

$$4r \sin(\theta) = (r \cos(\theta))^2 + (r \sin(\theta))^2$$

$$\frac{4r \sin(\theta)}{r} = \frac{r^2 \cos^2(\theta)}{r} + \frac{r^2 \sin^2(\theta)}{r}$$

$$\frac{4 \sin(\theta)}{r} = \frac{r \cos^2(\theta)}{r} + \frac{r \sin^2(\theta)}{r}$$

$$\frac{4 \sin(\theta)}{r} = \cos^2(\theta) + \sin^2(\theta) \Rightarrow \frac{4 \sin(\theta)}{r} = 1$$

identidade trigonométrica.

$$\boxed{r = 4 \sin(\theta)}$$

ALTERNATIVA A.

III Ache a equação cartesiana correspondente à: $r^2 = \cos \theta$.

$$r^2 = \cos(\theta) \quad : \quad r^4 = r^2 \cdot r^2$$

$$r^3 = r \cos(\theta) \quad r^2 \cos(\theta) = r \cdot r \cos(\theta) = r x$$

$$r^4 = r^2 \cos(\theta)$$

$$(x^2 + y^2)(x^2 + y^2) = r \cdot x$$

$$x = \frac{(x^2 + y^2)(x^2 + y^2)}{\sqrt{x^2 + y^2}} = \frac{(x^2 + y^2)^2}{\sqrt{x^2 + y^2}} = (x^2 + y^2)^{\frac{3}{2}} = x$$

$$x^2 = ((x^2 + y^2)^{\frac{3}{2}})^2 \Rightarrow \boxed{x^2 = (x^2 + y^2)^3}$$

ALTERNATIVA C.

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④ Integral que representa o comprimento da curva para

$$r = \sqrt{\theta} \text{ para } 0 \leq \theta \leq 9: \quad f(\theta) = (\sqrt{\theta})^2 = \theta$$

$$r = f(\theta) = \sqrt{\theta} \quad f'(\theta) = \frac{df(\sqrt{\theta})}{d\theta} = \theta^{\frac{1}{2}} = \frac{\theta^{\frac{1}{2}}}{2}$$

$$S = \int_0^9 \sqrt{f''(\theta)^2 + f'(\theta)^2} d\theta = \int_0^9 \sqrt{\frac{1}{4\theta} + \theta} d\theta \quad \left(\left(\frac{\theta}{2} \right)^{\frac{1}{2}} \right)^2 = \frac{\theta}{4}$$

$$\Rightarrow \int_0^9 \sqrt{\frac{1}{4\theta} + \frac{4\theta^2}{4\theta}} d\theta = \int_0^9 \sqrt{\frac{1+4\theta^2}{4\theta}} d\theta$$

ALTERNATIVA B.